Risk-based credibility assessment of two implicit FEA models for a coronary stent under fatigue and radial strength mechanisms per ASME V&V 40

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Abstract

Computational models of coronary stents are increasingly used to supplement bench evidence on endurance and scaffolding performance. We report a risk-based credibility assessment, organized per ASME V&V 40, for two implicit finite element analysis (FEA) models representing a balloon-expandable coronary stent: (1) a bare-stent lattice model computing element-level stress–strain histories under crimping, balloon expansion, and cyclic loading; and (2) a coupled stent–artery model computing response during diametral compression and cyclic bending. Three mechanisms of interest were evaluated: pulsatile fatigue driven by intraluminal pressure and curvature, bending-induced fatigue arising from cardiac motion, and radial crush strength under quasi-static compression. The context of use is virtual bench testing to supplement ISO-like durability and radial compression evidence for labeling; model risk is judged medium because simulation informs bench equivalence but is not sole basis for clinical deployment. Credibility activities were scoped to seven factors: numerical code verification, iteration and contact convergence behavior, spatial resolution effects, fidelity of parameter values and loads, agreement with bench measurements, applicability of the tests and measures to the stated use, and software quality practices. Both models passed analytic patch and manufactured solution checks within 0.1–0.3% of theory for linear benchmarks and within 1.2% for a Hertzian contact surrogate. Solver controls achieved sub-1e-6 force and energy residuals, with contact stabilization below 0.5% of peak contact force; path-following robustness was demonstrated across balloon expansion and crush sequences with continuation. Mesh convergence studies, including strut-thickness perturbations and contact discretization refinement, reduced hotspot strain-change below 2% between the two finest meshes for the lattice model and below 3% for the coupled model. Input parameters—geometry, material law, boundary conditions, and loading amplitudes—were traced to drawings, alloy certificates, and bench protocols; key uncertainties (e.g., cyclic hardening exponent, arterial wall modulus) were bounded by data and propagated to hotspot strain ranges. Comparisons with bench data included radial stiffness and crush strength versus mandrel-compression tests, and fatigue strain-range and life proxies derived from coupon S–N curves and stent samples instrumented with stereo-DIC in saline at 37 C. Using a Smith–Watson–Topper damage surrogate, median ratio of predicted to measured strain range was 0.96 (interquartile range 0.89–1.06) for pulsatile cycling of bare stents and 1.04 (0.93–1.12) for bending-induced hot spots in a surrogate vessel. Mapping of test articles, loading spectra, and measurement outputs to decision needs was documented, including traceable links to labeling claims and bench acceptance criteria. All activities were scored on a 0–5 Fitness-for-Decision scale for each factor across both models and all mechanisms. Numerical code verification scored 5 across all combinations. Spatial resolution effects scored 3–4; iteration and residual control scored 3–4; fidelity of parameter values and loads scored 3–4; agreement with bench measurements scored 3–4; and applicability of the evidence scored 3–5. Software quality assurance is summarized without a formal finding. The results delineate where the models are decision-ready and where targeted improvements—particularly in arterial property characterization and bending-mode comparators—would raise credibility for fatigue mechanisms while maintaining strong support for radial crush strength in the virtual bench role.

Keywords: coronary stent finite element analysis, ASME V&V 40 credibility, fatigue and radial strength

1. Introduction

Modern coronary stents are designed to balance deliverability, scaffolding, and durability in a dynamic vascular environment. The emergence of highly resolved finite element analysis (FEA) has enabled engineers to examine local stress–strain concentrations within complex stent geometries under physiologic and bench-representative loading. These insights inform design iteration, bench test planning, and, increasingly, labeling support via virtual bench evidence. However, the role of simulation in regulatory and clinical decision-making depends on how believable the model is for a stated purpose. ASME V&V 40 provides a framework for assessing that believability in a risk-informed manner. In this work we apply that framework to two implicit FEA models for a balloon-expandable coronary stent, focusing on pulsatile and bending-induced fatigue and on resistance to external radial compression.

The models differ in scope. One represents the bare stent lattice through crimping, balloon expansion, and cyclic loading without a vessel, supporting direct comparison to bench tests that operate on the isolated device. The other embeds the expanded stent in a deformable arterial wall, enabling assessment of contact states and load-sharing during crush and bending scenarios closer to in vivo mechanics. The context of use targets virtual bench testing as a supplement to ISO-like durability and radial compression programs. Therefore, the credibility questions we address include: How consistent are the codes' fundamental mechanics with analytic expectations? How tightly are iteration controls and contact enforcement managed in the nonlinear regimes pertinent to expansion and crush? How stable are results against mesh and discretization choices? How well are input parameters backed by traceable data and bounded uncertainties? How close are model outputs to bench-observed responses for the mechanisms of interest? And how well do validation activities connect to the decisions at stake in this context?

We report a comprehensive set of activities linked to these questions and score each on a 0–5 Fitness-for-Decision scale for every combination of model and mechanism. A large majority of the activities are common across models and tailored by mechanism where load cases and comparators differ. Our scope is intentionally restricted to seven credibility factors relevant to this role to maintain alignment with the decision to be supported and to manage the overall burden of evidence to be proportional to risk. While the paper does not make claims beyond fatigue and radial strength mechanisms, our process and findings offer a template for similar device-model assessments under ASME V&V 40.

Framing the risk-informed credibility question — gradation.

a.
Device, model, and decision use are stated qualitatively without explicit linkage.

b.
Device and model are specified; decisions are identified; mechanisms are named but not tied to outputs.

c.
Device, model(s), decisions, and mechanisms are mapped to concrete outputs; qualitative risk level is assigned.

d.
Device, model(s), decisions, and mechanisms are mapped to outputs and comparators; risk rationale is quantitative or semi-quantitative; burden of evidence is scoped.

e.
Full mapping from device to outputs to bench comparators with traceability; risk level and risk mitigation are justified; evidence burden is justified against risk and decisions.

2. Device and computational models (implicit FEA formulations and outputs)

The device is a balloon-expandable coronary stent intended to scaffold diseased coronary arteries. The open-cell lattice comprises repeating crowns connected by straight links in a helical pattern. Nominal expanded outer diameter is 3.0 mm; strut thickness is 90 μm; strut width is 80 μm; crown peak-to-peak span is 0.45 mm; and unit cell axial pitch is 1.2 mm. The base material is a Co–Cr alloy (L605) with room-temperature elastic modulus 230 GPa, Poisson's ratio 0.30, 0.2% offset yield strength 610 MPa, and ultimate tensile strength 1.1 GPa. Material characterization in saline at 37 C revealed a slightly reduced modulus at 225 GPa and cyclic stabilization over 100 cycles with a Ramberg–Osgood fit (K' = 1150 MPa, n' = 0.14). Low-cycle to high-cycle smooth-bar fatigue data were available from fully reversed and R = 0.1 tests at 20 Hz. These data underpin the damage proxies used in the models, as described later.

Model 1—the Stent Lattice Implicit FEA—represents the bare stent at three stages: crimping onto a rigid cylindrical mandrel to 1.2 mm OD, balloon expansion with pressure applied to the inner cell surfaces via a distributed shell-to-solid coupling to a rigid balloon surface, and cyclic loading to impose intraluminal pressure oscillation and superposed curvature. The mesh uses hexahedral elements (reduced integration with hourglass control) where feasible and wedge elements in geometric transitions. The final mesh for cyclic runs typically contains 1.1–1.5 million elements for a 12 mm-long stent. Contact between struts is modeled with a friction coefficient of 0.15 based on crimp-release experiments. Plasticity follows a temperature-corrected combined isotropic–kinematic hardening law calibrated to the stabilized hysteresis at 37 C.

Outputs of Model 1 include element-level von Mises stress, principal strains, plastic strain increments, and cycle-by-cycle strain ranges at hotspots in the crowns and link junctions. From these fields, we compute a Smith–Watson–Topper (SWT) parameter to correlate with life, acknowledging that this proxy is used in a relative sense to compare with bench fatigue indications. For balloon expansion and quasi-static compression, load–displacement curves are generated and benchmarked against bench data.

Model 2—the Stent-in-Artery Implicit FEA—adds a surrounding arterial wall represented as a thick-walled tube with three layers (intima-media-adventitia surrogate) and a neo-Hookean base response with an exponential fiber term to capture circumferential stiffening. At 37 C, the small-strain circumferential modulus ranges 0.5–1.2 MPa across layers; the exponential stiffening coefficients are tuned so that a pressurized, unstented 3.0 mm lumen exhibits 8–10% diametral compliance per 100 mmHg, consistent with proximal LAD measurements. The stent is expanded against the vessel to 3.0 mm lumen via balloon pressure coupling and contact with a friction coefficient of 0.05 to reflect lubricated saline conditions during deployment. After expansion and recoil, two load cases are applied: (a) diametral compression with rigid platens advancing to 1.5 mm radial displacement to measure crush resistance and contact state evolution; and (b) cyclic bending with combined curvature (±0.25 cm1 about a plane rotated 30 degrees from the crown plane) and pressurization (80–120 mmHg at 1 Hz) while both ends are constrained on sinusoidal axial motion tracks extracted from cineangiography of similar vessel segments.

Outputs of Model 2 include radial load–displacement response of the stent–artery system, maps of contact pressure and separation, and local strain ranges at stent hotspots influenced

Table 1: Nominal geometric and material parameters for the 3.0 mm coronary stent and arterial surrogate at 37 C.

Credibility factor	Level	Basis
Parameter	Level	One-line basis
Stent OD (expanded)	3.0 mm	Labeling size; matched to bench fixtures
Strut thickness	90 µm	Measured across 10 samples; 2 µm SD
Strut width	80 µm	Measured across 10 samples; 3 µm SD
Co–Cr elastic modulus	225 GPa	Saline, 37 C, ultrasonic pulse-echo
Yield strength (0.2% offset)	610 MPa	Vendor certificate; verified by tensile tests
Cyclic hardening (K', n')	1150 MPa, 0.14	Stabilized hysteresis fit
Arterial small-strain modulus	0.5–1.2 MPa	Layer-dependent; literature and in-house fit
Arterial compliance	8–10%/100 mmHg	Proximal LAD surrogate tuning

by vessel constraint. The artery model also reports vessel wall strain distributions to support a qualitative check that the model produces physiologically plausible distortions when bent and pressurized.

Both models employ an implicit quasi-static solver with full Newton iterations, line search, and augmented Lagrangian contact enforcement. Time integration is used as a load-stepping parameter only; inertial effects are suppressed except for numerical stabilization during severe contact transitions where a mass scaling factor of 1.0 (no scaling) or, if needed in sensitivity studies, 0.1 µs pseudo-time steps were tested to verify negligible kinetic energy relative to internal strain energy. The solution algorithms and control choices are detailed in the subsequent sections.

The selected combination of models enables two complementary perspectives: a bare-stent view that aligns directly with radial and fatigue bench tests conducted on the device alone, and a coupled view that exposes the role of vessel constraint and contact in altering local curvature and resistance to crushing. The pair addresses the mechanisms of interest from distinct but harmonized vantage points that together inform the intended virtual bench use.

3. Context of use: virtual bench testing to supplement ISO-like fatigue endurance and radial compression evidence for labeling; Model risk level: medium, because decisions inform bench equivalence and labeling but do not directly control clinical deployment. Rationale: an incorrect prediction could mischaracterize endurance or scaffolding claims, mitigated by mandatory bench testing that remains primary.

The intended role of the models is to complement—not replace—bench tests used to support labeling for fatigue endurance and radial compression. Specifically, the bare-stent model is used to examine how sensitive hotspot strain ranges are to realistic variations in pressure amplitude and curvature profiles within tolerance bands defined by bench protocols. It also predicts the load–displacement characteristic during diametral compression of the bare stent to size up differences between lots and to interpret failure modes seen in testing. The coupled stent–artery model is used to study whether vessel constraint alters locations and magnitudes of fatigue hot spots and to evaluate crush resistance pathways when external loads are applied with and without vessel support. These results, when viewed against bench outcomes, strengthen claims that the device meets endurance and scaffolding targets under the specified conditions.

We categorize model risk as medium for this application. Simulation results influence how bench programs are designed, how test parameter margins are justified, and how equivalence analyses are articulated in labeling. However, acceptance still requires primary bench tests, and no clinical deployment decisions are made solely from simulation outputs. Consequently, the burden of credibility evidence is calibrated to demonstrate that predictions are sufficiently reliable to guide bench interpretation and to support equivalence arguments. This calibration informed which factors were pursued and at what depth.

The mechanisms of interest and their comparators define the scope of model use. For fatigue, the emphasis is on strain-range predictions in stent crowns and links under pulsatile pressure and cardiac bending profiles that resemble in-bench cyclic tests in saline at 37 C. For radial strength, the focus is on quasi-static diametral compression tests employing rigid platens or mandrels with load measured against radial displacement. The two models span these circumstances, with the bare-stent simulation directly reflecting most bench fixtures and the coupled model adding insights about contact and constraint relevant to bending and crush in vivo, which helps interpret why bench outcomes cluster as they do.

Context of use clarity for a risk-informed assessment — gradation.

a.
Purpose, decisions, and boundaries are implied but not stated.

b.
Purpose and decisions are stated; boundaries are qualitative.

c.
Purpose, decisions, and boundaries are explicit; mechanisms and comparators are named.

d.
Purpose, decisions, boundaries, mechanisms, comparators, and stakeholders are explicit; risk level is articulated.

e.
All elements in the previous rung plus traceable links to labeling claims and bench acceptance criteria; rationale for evidence depth tied to risk mitigation.

Figure 1: Schematic of the two finite element models and load cases: (a) bare stent crimping, expansion, and cyclic pressure–curvature; (b) stent expanded in a three-layer arterial tube undergoing diametral compression and cyclic bending.

4. Mechanisms of interest and their experimental comparators

Pulsatile fatigue is driven by cyclic intraluminal pressure changes that expand and contract the stent–artery lumen, superposed with curvature changes due to cardiac motion. In bench tests, these are emulated using saline at 37 C with pressure cycles between 80 and 120 mmHg at 1–2 Hz, sometimes augmented by a controlled curvature amplitude. We use two such configurations: (1) bare stent in a compliant saline-filled tube with pressure cycling and minimal curvature, to isolate pressure-driven strains; and (2) bare stent or stent-in-artery placed on a bending apparatus with a set curvature applied about a rotated plane, while pressure cycles operate concurrently. Digital image correlation (DIC) with stereo microscopes captures surface strain proxies at crowns and links; additionally, low-noise strain gauges on micromachined coupons made from strut stock underpin the S–N relationship extrapolated to the stent geometry through SWT correlation.

Bending-induced fatigue emphasizes the role of curvature that changes the local geometry of crowns and link junctions, amplifying notch effects. The comparator uses a programmable bending platform in heated saline with curvature amplitude ±0.20–0.30 cm1 at 1 Hz and synchronized pressure oscillations. The bending plane is offset by 30 degrees relative to the crown plane to generate asymmetric strain fields similar to those captured by high-speed angiography of stented segments. End constraints emulate wire snaring conditions by constraining axial displacement within ±5% of length change over the cycle to represent tethering from neighboring tissue. DIC measurement uncertainty is 80–120 microstrain after filtering; cycle tracking identifies peak and valley frames across 100 stabilized cycles.

Radial crush strength is assessed by quasi-static diametral compression using rigid platens or mandrels per internal protocols consistent with ISO 25539-2 intent for balloon-expandable stents. For the bare device comparator, one cycle of loading to 2.0 mm radial displacement and unloading provides the load–displacement curve, collapse onset, and residual deformation. The stent-in-artery comparator uses a saline-inflated, compliant gelatin vessel analog with wall stiffness matched to the arterial surrogate, allowing a qualitative check of contact engagement and a quantitative comparison for overall load support when the vessel is present. Force is measured by a 50 N load cell with 0.1 N resolution; displacement is recorded by an LVDT with 5 µm resolution. Five samples per condition are tested.

Across all mechanisms, we align measurement outputs with model outputs: load vs displacement for crush; local peak–to–valley strain range at crowns and links for fatigue. For fatigue comparators where direct stent life testing is not performed to failure, we use an SWT-based proxy: $\Delta\epsilon a$ cycles and mean stress-corrected measure, combined with coupon S–N fits, to estimate relative life indices for matching stent locations. While life prediction in absolute cycles is not claimed, the relative ranking and magnitude of strain ranges feed directly into bench equivalence reasoning and margin analyses, which is the intended role for the virtual bench models.

Definition of mechanisms and comparators — gradation.

a.
Mechanisms are identified but not operationalized; comparators are vague.

b.
Mechanisms are named with qualitative comparators.

c.
Mechanisms have quantitative comparator conditions; outputs are aligned at a high level.

d.
Mechanisms have quantitative comparator conditions with measurement uncertainty; outputs are mapped to model fields.

e.
Mechanisms and comparators are fully specified with traceability to protocols and instrumentation; uncertainty budgets are stated; model-to-measurement mapping is explicit.

5. Credibility plan constrained to seven factors: Discretization error; Model inputs; Numerical code verification; Numerical solver error; Output comparison; Relevance of the validation activities to the COU; Software quality assurance

The credibility plan is scoped to the seven factors most critical to the role of these models in virtual bench support. The plan allocates depth proportionally to how each factor contributes to confidence for the mechanisms studied. For

Table 2: Bench comparator conditions by mechanism.

Credibility factor	Level	Basis
Mechanism	Level	One-line basis
Pulsatile fatigue	80–120 mmHg at 1–2 Hz; saline 37 C; optional curvature ±0.10 cm1	Pressure-driven strain ranges in crowns and links; stereo-DIC
Bending-induced fatigue	Curvature ±0.20–0.30 cm1; pressure 80–120 mmHg; 1 Hz	Amplified local curvature; stereo-DIC at hotspots
Radial crush strength	Platens to 1.5–2.0 mm radial displacement; 0.1 N force resolution	Load–displacement curve; collapse onset; residual set

instance, code-level checks are asked to be exhaustive given their cross-cutting importance, while relevance mapping focuses on traceability between bench comparators and the outputs used to support labeling. Each activity produces objective evidence (e.g., error norms, residual histories, convergence rates, quantitative comparisons to tests) and culminates in a 0–5 score per factor for each combination of model and mechanism. The Fitness-for-Decision scale is defined as: 0, no usable evidence; 1, limited qualitative support; 2, partial quantitative support with significant gaps; 3, moderate support with acknowledged limitations; 4, strong support with minor limitations; 5, decision-grade evidence with well-bounded limitations.

While some factors might be tackled differently for the two models, we apply common templates to ensure comparability. Discretization activities use h-refinement, strut-thickness perturbation, and contact discretization sensitivity. Parameter fidelity reviews cover geometry, material models, boundary conditions, and load amplitudes with traceability and uncertainty bounds. Iteration and contact checks document residual tolerances, convergence behavior, and path-following robustness. Agreement with measurements is evaluated against radial compression data and fatigue strain-range proxies from coupons and stents. Applicability of evidence is addressed by mapping comparators and outputs to the context of use. Practices around software versioning, configuration, and runtime logging are summarized.

6. Numerical code verification (analytic patch tests and manufactured solutions; both models)

We performed code-centric checks to demonstrate that the implementation of continuum mechanics, contact enforcement, and plasticity in the analysis tool reproduces known solutions with tight error. In linear elasticity, a 3D cantilever beam subject to an end moment was meshed with second-order hexahedral elements to avoid shear locking; the analytic tip rotation and stress distribution were recovered with errors below 0.12% in energy norm at six elements through thickness, decreasing to 0.03% at ten elements. A pure bending patch in a thin plate confirmed plane-stress kinematics and stress-free thickness directions to within 0.05% of analytic solutions when reduced integration with hourglass control was used in solid elements, validating the specific element technology applied to struts.

For contact, a Hertzian cylinder-on-plane surrogate with known load–area scaling was constructed using the same augmented Lagrangian algorithm as the stent models. Normal pressure integrated over contact matched the theoretical value to within 1.2% at the finest surface mesh; penalty and augmentation parameters were varied over an order of magnitude without instability, demonstrating robustness of the enforcement scheme in the contact pressure range (2–12 MPa) encountered during crimping and crush.

Plasticity and kinematic hardening were checked using a uniaxial cyclic loading of a representative material point with the L605 parameters at 37 C. Stabilized hysteresis loop area and mean stress evolution over 50 cycles matched the calibrating curve to within 0.4% in energy dissipation and 1.6 MPa in mean stress for the default return mapping tolerances (1e-8). An axisymmetric notched bar MMS (manufactured solution) introduced a known spatially varying plastic strain field through an imposed body force; recovered equivalent plastic strain deviated by less than 0.6% in L2 norm at the targeted refinement.

Collectively these checks, executed on the same solver version and element formulations used in the stent models, demonstrate that the underlying numerical machinery produces the right answers for simplified problems spanning the relevant physics: linear elasticity, contact mechanics, and cyclic plasticity. The tests were repeated after minor code updates to ensure no regressions; changes in error metrics remained within the natural scatter of discretization when mesh density was held constant.

Numerical code verification — gradation.

a.
No formal verification; reliance on vendor statements.

b.
Basic static elasticity patch tests run; qualitative agreement documented.

c.
Multiple analytic checks across elasticity and contact; quantitative errors reported.

d.
Analytic and manufactured-solution checks across elasticity, contact, and plasticity; tolerances set; regression monitored.

e.
Comprehensive MMS- and analytic-based verification across all governing equations and algorithms used; error norms within pre-specified thresholds; automated regression controls in place.

7. Numerical solver error (residual tolerances, contact convergence, path-following robustness; per model × mechanism)

Solver controls were established to ensure residuals, contact enforcement, and load-path tracking were tight enough that iteration artifacts were not a dominant error source for any mechanism. For both models, full Newton iterations with line search were used. Force and energy residual tolerances were set to 1e-6 relative to the total applied nominal load and internal energy increments, respectively. For contact, the augmented Lagrangian penalty was set to achieve less than 0.5% penetration normalized by local element size at peak contact pressure, with augmentation updated every other iteration until complementarity residuals fell below 1e-5 normalized units. For large-deflection steps such as balloon expansion and early crush, arc-length continuation stabilized progress through snap-through-like phenomena; arc-length constraints were tightened adaptively when the slope of the load–displacement curve softened.

In Model 1 under pulsatile cycling, the pseudo-time increment per cycle was set so that three substeps spanned peak–valley–peak; each substep converged within 4–7 iterations typically, with outliers at 9 iterations; across 100 cycles used for stabilization and measurement, 0.8% of substeps required line search relaxation. The ratio of kinetic to internal energy remained below 0.01 throughout, indicating essentially quasi-static behavior. For bending-induced cycling, where curvature induces asymmetric contact states during recoil, one out of six runs required a local contact stabilization ramp that added 0.2% to the peak contact penalty before converging cleanly.

In Model 2 under diametral compression, the presence of the vessel adds compliance and contact path multiplicity. The solver converged at each displacement increment of 0.05 mm within 5–10 iterations; near collapse onset of the stent crowns, arc-length continuation was triggered automatically and maintained equilibrium through the flattening of the crown segment. Comparison of runs with and without arc-length showed less than 0.3% difference in the recorded load at 1.5 mm radial displacement, suggesting that the continuation method did not distort the force response. For cyclic bending, convergence was more variable: in two instances among eight runs, local contact loss–regain at link junctions led to transient increases in iteration counts to 12–14, after which convergence resumed under the same tolerances. The final stabilized strain ranges differed by 0.5–0.8% between repeated runs with randomized initial arc-length seeds, indicating that iteration path variability is small compared to other sources.

Residual contour plots and contact gap distributions were archived for all runs serving as evidence here. The overall behavior indicates that iteration errors are contained through a combination of stringent tolerances, adaptive continuation, and, when needed, mild contact stabilization. These practices were held constant across models and mechanisms except where sensitivity checks guided slight tightening.

Iteration and contact convergence rigor — gradation.

a.
Default solver settings used; no monitoring.

b.
Basic convergence logs reviewed; tolerances at defaults.

c.
Tolerances set; residuals and iteration counts monitored; contact parameters justified.

d.
Tolerances and contact parameters tuned per load regime; arc-length/continuation applied across softening; energy measures monitored.

e.
All previous plus repeated runs for stability; documented sensitivity of outputs to tolerances; convergence artifacts shown to be negligible relative to other uncertainties.

$$\text{CI\_\{score\}} = \frac{\sum_{i=1}^{n} w_i\, x_i}{\sum_{i=1}^{n} w_i} \tag{1}$$

$$\text{\textbackslash varepsilon\_\{rel\}} = \frac{\sum_{i=1}^{n} w_i\, x_i}{\sum_{i=1}^{n} w_i} \tag{2}$$

$$\text{U\_\{val\}} = \frac{\sum_{i=1}^{n} w_i\, x_i}{\sum_{i=1}^{n} w_i} \tag{3}$$

8. Discretization error (systematic mesh refinement, strut-thickness sensitivity, contact discretization; per model × mechanism)

Spatial resolution effects were probed with a structured plan of h-refinement and contact surface discretization, combined with a strut-thickness perturbation intended to reveal sensitivity to potential manufacturing variability. For Model 1, three meshes were constructed for a 12 mm-long stent: coarse (0.020–0.025 mm element edge in crowns), medium (0.012–0.014 mm), and fine (0.006–0.008 mm). Hotspot strain range at the crown apex under 80–120 mmHg cycling changed by 5.8% from coarse to medium and by 1.6% from medium to fine. The link–crown junction showed 6.4% and 1.9% changes over those same refinements. Extrapolation with a Richardson-like approach suggested an asymptotic error of approximately 1.1% at the fine mesh relative to the numerical zero-mesh-limit estimate for these measures. For bending-induced cycling, where curvature is more pronounced, the coarse-to-medium change was 7.5% and the medium-to-fine change was 2.3% at the dominant crown hotspot.

Contact discretization was examined by refining the inner surface mesh used to distribute balloon pressure and to handle strut–strut contact during crimping. Doubling the surface facet density reduced the variance of local contact pressure by 12% and altered post-expansion recoil by 0.4%, which propagated to a 0.6% change in average crown curvature—small compared with other mesh effects. Strut-thickness perturbations of ±5 μm (within observed manufacturing scatter) changed peak strain range by +3.2% and 2.8% respectively for pulsatile cycling at the fine mesh.

For Model 2, the coupled stent–artery system introduces additional discretization considerations in the vessel wall. Three vessel meshes were tested: 12, 24, and 36 elements through the wall thickness with graded bias towards the intimal surface; the stent mesh refinement mirrored the Model 1 progression. Under diametral compression to 1.5 mm radial displacement, the predicted load changed by 4.1% from coarse to medium and 1.2% from medium to fine. Hotspot strain range under cyclic bending changed by 6.9% and 2.6% across the same refinements at a crown in contact with the vessel. Contact tracking between stent and vessel remained stable with a facet-to-element ratio above 3:1, and refining the vessel surface beyond that brought less than 0.5% change in hotspot strain range.

Collectively, these observations support that the two finest meshes capture the primary spatial gradients in the stent and contact interfaces for the mechanisms examined. Residual differences at the fine mesh level are small enough to treat discretization effects as a minor contributor to overall uncertainty in the cyclic strain ranges and crush loads presented.
Spatial resolution adequacy — gradation.

a.
Single mesh used; no refinement evidence.

b.
Coarse and medium meshes compared qualitatively.

c.
At least three meshes compared with quantitative change in key outputs.

d.
Three or more meshes plus contact surface refinement and geometry perturbations; extrapolation used to estimate numerical error.

e.
Comprehensive space–time and contact discretization studies with quantified numerical error bounds relative to decision thresholds.

9. Model inputs (geometry, material law, boundary conditions, loading amplitudes; traceability and uncertainty bounds; per model × mechanism)

Geometry: Lattice geometries were reconstructed from CAD with verification against micro-CT of five stents per lot. For the 3.0 mm device, mean strut thickness was 89.8 μm (SD 2.1 μm) and mean strut width was 80.7 μm (SD 3.3 μm). Crown radii measured 78 μm (SD 5.4 μm). These distributions informed the ±5 μm strut-thickness perturbations used in discretization sensitivity and the crown curvature initial conditions for the expanded shape.

Material law: The Co–Cr cyclic behavior at 37 C was characterized on smooth-bar specimens machined from the same lot as the stent stock. Fully reversed (R = 1) and R = 0.1 tests at 20 Hz established a stabilized response by 100 cycles. The combined hardening law parameters were obtained by least-squares to hysteresis loops at strain amplitudes 0.2%, 0.4%, and 0.6%. The SWT life correlation was built using coupon S–N data: Basquin slope b = 0.082 (R = 0.1) with C = 980 MPa, and Morrow mean stress correction for the mean stress observed in stent locations. Scatter in coupon fatigue life produced a 95% confidence band corresponding to ±13% in equivalent strain amplitude at 10e7 cycles.

Boundary conditions and loads: For pulsatile cycling, pressure amplitude was set to 80–120 mmHg and curvature to ±0.10 cm1 nominally; sensitivity runs explored ±10 mmHg and ±0.05 cm1 variations. For bending-induced cycling, curvature was ±0.25 cm1 nominal with a 30-degree rotated plane; axial end constraints allowed ±5% axial displacement relative to free bending. For diametral compression, platens advanced to 1.5 or 2.0 mm radial displacement at 0.5 mm/min. Friction coefficients were assigned as 0.15 for strut–strut contact (crimp) and 0.05 for stent–artery contact (saline-lubricated). Vessel material parameters were tuned to compliance targets derived from literature and in-house pressurization tests on harvested porcine coronary segments sized to similar diameters.

Uncertainty bounds and propagation: Geometry and material uncertainties were propagated through a local sensitivity analysis. For the lattice model pulsatile case, ±2 μm in thickness and ±0.02 in cyclic hardening exponent n' produced ±1.7% and ±1.3% change, respectively, in dominant crown strain range. For the coupled model bending case, ±0.2 MPa in circumferential wall modulus altered hotspot strain range by ±2.1%. Pressure and curvature uncertainty bands yielded ±1.2% and ±3.9% changes in hotspot strain range respectively. These were combined in quadrature as an indicative bound on input-driven variability for context when comparing to bench observations.

Traceability: All input values are documented against sources: drawings and CT metrology for geometry; tensile and cyclic coupon tests for stent alloy; vessel surrogate calibration logs; crimp and expansion bench observations for friction coefficients; and bench protocol specifications for loads. Where literature provided ranges (e.g., coronary compliance), we report the range and the selection rationale to align with comparator conditions.
Credibility of parameter values and loads — gradation.

a.
Inputs taken from nominal design; no traceability.

b.
Inputs traceable to at least one data source; uncertainties not quantified.

c.
Inputs traceable to multiple sources; key uncertainties bounded qualitatively.

d.
Inputs traceable with quantitative uncertainty bounds; sensitivity of outputs to key inputs quantified.

e.
All previous plus systematic uncertainty propagation to decision metrics; rationale for selected values tied to comparator conditions.

10. Output comparison (bench-referenced radial stiffness and crush strength; fatigue strain-range and life proxies vs coupons and stent samples; per model × mechanism)

Radial crush strength: For the bare stent, mandrel compression to 1.5 mm radial displacement produced a mean load of 9.8 N (SD 0.6 N) in bench tests (n = 5). Model 1 predicted 10.1 N at the same displacement, a +3.1% deviation. The initial slope of the load–displacement curve over the first 0.3 mm was 16.5 N/mm in bench and 16.1 N/mm in simulation (2.4%). Collapse onset—the inflection indicating crown snap-through—occurred at 1.25 mm in bench and at 1.23 mm in simulation. Residual set after unloading to zero load was 0.12 mm in bench and 0.11 mm in simulation. For the stent–artery system, bench data using a compliant gelatin tube indicated a mean load of 11.4 N (SD 0.7 N) at 1.5 mm compression; Model 2 predicted 11.0 N (3.5%). The presence of the vessel shifted collapse onset later by 0.08 mm in both experiment and simulation, and contact pressure maps confirmed that vessel engagement redistributed loads to adjacent crowns as seen in high-resolution fluoroscopy.

Pulsatile fatigue strain ranges: For bare stents in a pressure-cycling setup, stereo-DIC measured peak–to–valley axial strain range at the dominant crown hotspot as 0.34% (median; IQR 0.31–0.37%; n = 6 devices). Model 1 predicted 0.33% (median across 100 stabilized cycles), a ratio of 0.97. At a link–crown junction, DIC measured 0.22% (IQR 0.20–0.24%), while the model predicted 0.21% (ratio 0.95). Translating to SWT equivalent, the median predicted-to-measured ratio across hotspots was 0.96 (IQR 0.89–1.06). Using coupon S–N curves with Morrow correction, the relative life index at the dominant crown was within +8% of the bench-inferred index, and locations ranked consistently between model and measurement in 10 of 12 comparisons.

Bending-induced fatigue strain ranges: In the bending platform at ±0.25 cm1 and 80–120 mmHg, stereo-DIC measured a dominant crown hotspot strain range of 0.46% (IQR 0.42–0.49%; n = 5 devices). Model 1, which does not include a vessel, predicted 0.43% for the same curvature and pressure cycle in air, a ratio of 0.93; including a soft support tube in the bench narrows this gap to 0.95 as the boundary condition approaches the lattice model assumption. Model 2 with a vessel predicted 0.48% at the same location, a ratio of 1.04; the model shifted the absolute location of the hotspot by roughly 60 μm circumferentially due to vessel constraint, which matched the observed offset within the DIC spatial resolution limit. Across measured hotspots, the median predicted-to-measured strain-range ratio was 1.04 (IQR 0.93–1.12) for Model 2.

Agreement consistency and uncertainty: For all comparisons, we included measurement uncertainty bands and input uncertainty propagation bounds. The radial crush deviations fall well within the combined uncertainty of bench load calibration (±1%) and compliance of fixtures (±1–2% in displacement under load). For fatigue strain ranges, the DIC uncertainty (80–120 microstrain) corresponds to ±0.02–0.03% strain range at the measured magnitudes, while input-induced variabilities add ±0.02–0.04%. The observed model-to-measurement ratios are therefore within expected bounds given the combined experimental and input uncertainties. The consistency of ranking in fatigue hotspots between model and measurement supports the intended use to interpret relative severity rather than predict absolute life.

Output comparison — gradation.

a.
Qualitative comparison to limited data; no uncertainty accounting.

b.
Quantitative comparison to at least one dataset; limited uncertainty discussion.

c.
Multiple datasets used; key outputs compared with uncertainty bands; ratios or errors reported.

d.
Multiple datasets across mechanisms; spatial field comparisons at hotspots; combined uncertainty considered.

e.
Systematic, mechanism-wise comparisons with traceable protocols; uncertainty budgets propagated; acceptance criteria established relative to decision needs.

11. Relevance of the validation activities to the COU (mapping test articles, loading spectra, and measurement outputs to decision needs; per model × mechanism)

The virtual bench role demands that comparator conditions and measured quantities align with what the simulation predicts and what supports labeling. For radial crush, Model 1 predicts the bare stent load–displacement characteristic under diametral compression with rigid platens—the same as in bench testing used for labeling. The model's output, including collapse onset and residual set, maps directly to bench acceptance criteria and to scaffold claims. Model 2, while including a vessel, informs interpretive questions about how in vivo support may increase effective stiffness and alter collapse modes; this is relevant for understanding performance margins but is less directly tied to labeling tests.

For fatigue, Model 1's pulsatile-pressure scenarios using the bare lattice reflect the bench configurations where direct stent measurements are made in heated saline at specified pressure cycles, often with minimal curvature. The model's hotspot strain ranges map to DIC-measured ranges and the SWT proxy connects to coupon S–N fits used to reason about endurance margins. For bending-induced fatigue, the addition of a vessel in Model 2 more closely mimics in vivo curvature amplification and contact constraints; the mapping of hotspot locations and magnitudes to DIC observations is strengthened in this case, rendering Model 2 the more applicable tool for bending scenarios, while Model 1 still offers value for bounding and sensitivity studies under simplified boundary conditions.

Documented traceability ties each comparator to a decision element: crush compliance to scaffolding claims and size labeling; pulsatile strain range magnitudes to endurance equivalence between lots and designs; and bending hot spot patterns to rationales about areas at greater risk under motion. We maintain a matrix linking model outputs to comparator measures and to labeling statements, ensuring that the evidence gathered bears directly on the questions asked in the intended context.

Applicability of evidence to the intended role — gradation.

a.
Comparators are described but their connection to the intended use is assumed.

b.
Comparators are linked qualitatively to model outputs.

c.
Comparators and outputs are mapped with examples; some traceability to decisions.

d.
Comparators and outputs are mapped comprehensively with traceability to decisions and acceptance criteria.

e.
Evidence mapping is complete and prioritized to the intended role; applicability gaps are identified and bounded.

12. Software quality assurance (toolchain description, change control, and run-time logs noted briefly)

Analyses were executed in a commercial implicit nonlinear FEA solver (version 2025.1) with user subroutines for cyclic plasticity written in C++ and compiled with GCC 12.2 under Linux. Pre- and post-processing used in-house scripts version-controlled with Git; solver decks and results were archived with metadata including solver version, hardware identifiers, and checksums. A change-control log documents updates to the plasticity subroutine and contact parameter defaults; unit tests for the subroutine were run automatically on commit.

Run-time logs for all verification, discretization, and comparator simulations were retained, including residual histories, iteration counts, and warnings. Hardware was a 32-core workstation with ECC memory; no soft errors were detected by ECC in the reported runs. Solver vendor patches during the study window were reviewed for impact on contact enforcement and plasticity routines; verification cases were re-run after one minor patch with no change in error metrics outside mesh-discretization scatter.

Licensing records and environment modules were maintained to ensure reproducibility of the solver environment. A regression suite was run monthly on the verification cases to detect inadvertent drift from code or environment changes.
Software quality assurance practices — gradation.

a.
Ad hoc tool use; no records.

b.
Version information recorded; changes noted informally.

c.
Version control for scripts; change logs for critical routines; run logs retained.

d.
Version control, change control, and automated unit tests for custom code; environment and solver versions documented; regression runs on verification cases.

e.
Formal quality system for software lifecycle; automated reproducibility; independent code reviews; audit trail linking runs to decisions.

13. Credibility results matrix (0–5 Fitness-for-Decision scores reported for each factor across both models and all three mechanisms, with short justifications per cell)

We scored each factor on the 0–5 Fitness-for-Decision scale for both models and all mechanisms, grounding each score in the quantitative evidence presented. The table lists each model × mechanism × factor with the assigned level and a one-line basis. Scores reflect the strength of evidence relative to the intended role; they do not imply suitability for other contexts or for clinical decision-making beyond virtual bench support.

14. Discussion (implications for labeling support and model use boundaries within the stated COU)

The paired-model strategy proved useful for the intended virtual bench role. For crush, the bare-stent lattice model's direct alignment with bench fixtures yielded strong agreement on load–displacement behaviors, instilling confidence in its use to frame equivalence discussions and to probe margins. The coupled stent–artery model added mechanistic clarity by showing how vessel support alters collapse pathways, providing context to why observed bench loads increase with a soft sleeve. For fatigue, the lattice model provided credible strain-range magnitudes under pressure-only cycling that are consistent with DIC measurements; it was especially informative for studying sensitivity to pressure and curvature bounds defined by bench protocols. The coupled model improved the mapping in bending conditions by shifting hotspots to anatomically plausible locations and by increasing strain ranges where the vessel constrains crown flattening, yielding closer correspondence to DIC patterns in bending benches.

We also delineated practical boundaries. First, these models are suited to predicting local cyclic strain ranges and relative life indices using SWT surrogates, not absolute life in cycles to fracture. Second, model-to-bench matches depend on faithfully reproducing boundary conditions—particularly curvature magnitude and end constraints for bending scenarios. Third, the artery representation is an idealized, layered tube; its properties are tuned to compliance ranges relevant to bench analogs, and it should not be over-interpreted as a patient-specific model. Within these boundaries, the credibility activities provide a robust foundation for the stated role.

Findings for the factor addressing mesh-induced uncertainty are as follows: across all combinations, the lowest and highest levels were 3 and 4, respectively, with an average of approximately 3.5 driven by 1.2–2.6% change between the two finest meshes at hotspots or crush loads, and by small effects of contact surface refinement.

For the factor on fidelity of parameter values and loads, scores ranged from 3 to 4 over all combinations, averaging about 3.2, reflecting traceability to drawings and tests, bounded uncertainties (e.g., ±2 μm thickness, ±0.02 in hardening exponent, ±0.2 MPa in vessel modulus), and quantified sensitivity causing roughly ±1–4% changes in hotspot strain or crush response.

For Numerical code verification, the assigned level was 5 consistently, supported by error norms within 0.1–1.2% for elasticity and contact surrogates and within 0.6% for plastic MMS, with regression checks confirming stability across minor solver updates.

Regarding iteration and residual control, levels spanned 3 to 4 with a mean near 3.7, based on tight residual tolerances

(1e-6), robust arc-length continuation through softening, and only occasional increases in iteration counts during contact loss–regain without measurable impact on stabilized strain ranges or crush forces.

For Output comparison, levels fell between 3 and 4 with an average around 3.2, mirroring median predicted-to-measured ratios of about 0.96 for pulsatile cases and 1.04 for bending in the coupled model, and crush loads within roughly ±3–4% of bench values; all within combined experimental and input-induced uncertainty bands.

Finally, for how well the tests and measures align with the decisions in scope, levels ranged from 3 to 5 with average about 4.2, highest where the model output maps directly to labeling measurements (bare-stent crush) and strong where the coupled model reproduces bending hotspots relevant to fatigue rationale; lower where coupled crush informs interpretation rather than labeling directly.

The scores and their bases recommend targeted next steps. Improving characterization of arterial material properties in the bending range and their variability would raise the input-fidelity and discretization confidence for the coupled model. Expanding bending comparator datasets with higher-resolution DIC and varied curvature planes would further strengthen the agreement factor. Automating mesh adaptivity at hotspots could reduce residual discretization effects. On the solver side, exploring penalty adaptation and contact friction regularization may suppress the occasional iteration count spikes seen during contact transitions. None of these improvements are expected to materially alter the strong position for crush in the virtual bench role, but they would lift the fatigue mechanisms towards higher levels in the medium-risk setting.

From a decision-making standpoint, the present evidence supports using Model 1 to: (1) predict bare-stent crush loads and stiffness within a few percent of bench values; (2) estimate pulsatile hotspot strain ranges to guide endurance margins and equivalence framing; and (3) rank bending hotspots under simplified curvature conditions when comparator benches reflect such constraints. Model 2 is supported to: (1) evaluate how vessel constraint shifts and amplifies bending hotspots, to bolster fatigue rationales; and (2) interpret why crush resistance differs with sleeve support, complementing bare-stent benches. Both models should be applied within the defined ranges of pressure, curvature, displacement, and material properties documented here. Outside these ranges, new credibility activities would be needed before extending claims.

15. Limitations (scope restricted to seven credibility factors; no claims beyond fatigue and radial strength)

This assessment is intentionally scoped to seven credibility factors keyed to the virtual bench role for fatigue and radial strength. Other factors that sometimes appear in broader credibility programs are outside the present scope. The fatigue evidence relies on strain-range proxies and coupon S–N correlations; absolute life in cycles to fracture is not claimed. Vessel properties are represented by an idealized layered tube tuned to compliance targets rather than patient-specific anatomy; therefore, inferences about in vivo outcomes are limited to mechanism-level interpretations, not clinical prediction. Bending comparators, while carefully executed, are finite in number and span; adding planes and amplitudes could further challenge the models. Finally, all activities are tied to the specific 3.0 mm device geometry and material; extrapolation to different designs or sizes would require reapplication of the same credibility process.

16. Conclusions and next steps (focused improvements for low-scoring factor–mechanism pairs)

We conducted a risk-based credibility assessment per ASME V&V 40 for two implicit FEA models of a balloon-expandable coronary stent under pulsatile and bending-induced fatigue and under radial crush strength mechanisms. Numerical code verification achieved decision-grade confidence across elasticity, contact, and cyclic plasticity. Iteration controls and contact convergence were well managed, with rare transients. Spatial resolution effects were contained to small residuals at fine meshes; input fidelity was strong where geometric and material data were directly measured and bounded, with recognized areas for improvement in vessel properties. Agreement with measurements supported the virtual bench role: crush loads and stiffness matched within a few percent; fatigue strain-range proxies aligned with DIC to within uncertainty bands, with the coupled model providing the closest mapping for bending. Applicability of the validation activities to the intended role was strongest where the model mirrored bench fixtures and was still robust where it complemented bench observations for interpretive use.

Improvements targeting the lower-scoring pairs include: (1) expanding the arterial property database under bending strains and frequencies typical of coronary motion to reduce input uncertainty in the coupled model; (2) adding bending comparators with varied planes and amplitudes to enrich output comparisons; (3) adopting automated mesh adaptivity at hotspots to reduce residual discretization effects, especially under bending; and (4) refining contact regularization to further stabilize iterations during transient loss–regain events. These steps would raise the levels for bending-induced fatigue factors while maintaining strong support for crush. We will also continue routine verification regression to secure the code-level foundation.

Within the present medium-risk context, the models are ready to support virtual bench tasks: interpreting and bounding fatigue strain ranges for endurance rationale and predicting crush performance to back scaffolding claims. As we complete the proposed improvements, we anticipate updating the results matrix, seeking to move the bending-related discretization and input-fidelity factors closer to level 4 and to confirm, with expanded comparators, that output comparison for fatigue can be elevated accordingly.

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Table 3: Fitness-for-Decision scores (0–5) by model × mechanism × factor with one-line justification.

Credibility factor	Level	Basis
Model 1 – Pulsatile fatigue – Numerical code verification	5	Elasticity, contact, and plasticity checks within 0.1–1.2% of theory; regression stable
Model 1 – Pulsatile fatigue – Numerical solver error	4	Residuals ≤1e-6; 4–7 iterations typical; negligible kinetic energy; rare line search
Model 1 – Pulsatile fatigue – Discretization	4	Medium-to-fine change ≤1.9% in hotspot strain; contact facet refinement effect ≤0.6%
Model 1 – Pulsatile fatigue – Model inputs	3	Inputs traceable; key uncertainties bounded; sensitivity shows ±1–4% effect on strain
Model 1 – Pulsatile fatigue – Output comparison	3	Median strain-range ratio 0.96; IQR within DIC and input uncertainty bands
Model 1 – Pulsatile fatigue – Relevance to COU	4	Direct mapping to bare-stent pressure-cycling bench; outputs align with DIC and SWT
Model 1 – Pulsatile fatigue – Software quality assurance	3	Version control and logs in place; unit tests for custom code; no formal audit
Model 1 – Bending-induced fatigue – Numerical code verification	5	Same verified algorithms; applicable to bending with contact
Model 1 – Bending-induced fatigue – Numerical solver error	4	Residuals controlled; one of six runs needed mild contact stabilization
Model 1 – Bending-induced fatigue – Discretization	4	Medium-to-fine change 2.3% at hotspot; thickness perturbation ±3% effect
Model 1 – Bending-induced fatigue – Model inputs	3	Curvature and end constraint bounds traced; sensitivity ±3–4% on strain
Model 1 – Bending-induced fatigue – Output comparison	3	Predicted-to-measured ratio 0.93–0.95; location match within DIC resolution
Model 1 – Bending-induced fatigue – Relevance to COU	4	Maps to simplified bending benches; supports sensitivity and ranking in virtual bench
Model 1 – Bending-induced fatigue – Software quality assurance	3	Toolchain controlled; regression on verification cases
Model 1 – Radial crush strength – Numerical code verification	5	Contact and plasticity verified; load–area and elastic checks precise
Model 1 – Radial crush strength – Numerical solver error	4	Arc-length stabilized collapse; ≤0.3% effect on 1.5 mm load
Model 1 – Radial crush strength – Discretization	4	Medium-to-fine change 1.2% in 1.5 mm load; contact refinement <0.5%
Model 1 – Radial crush strength – Model inputs	4	Geometry and friction well characterized; low sensitivity of load to inputs
Model 1 – Radial crush strength – Output comparison	4	Load at 1.5 mm within +3.1%; slope within 2.4%; collapse onset within 0.02 mm
Model 1 – Radial	5	Directly mirrors labeling

Table 4: Summary of factor-level findings used in decision-making, restating numbers given in the prose.

Credibility factor	Level	Basis
Spatial resolution effects (Discretization)	Min 3; Max 4; Mean 3.5	Two-finest-mesh change 1 2.6% at hotspots or cru loads
Fidelity of parameter values and loads (Model inputs)	Min 3; Max 4; Mean 3.2	Traceable data; sensitivit yield ±1–4% changes in k outputs
Numerical code verification	All 5	Elasticity/contact/plasticit verification within 0.1–1. (0.6% MMS)
Iteration and residual control (Numerical solver error)	Min 3; Max 4; Mean 3.7	Tolerances 1e-6; robust c tinuation; rare iteration spi
Output comparison	Min 3; Max 4; Mean 3.2	Strain-range ratios 0.96–1. crush within ±3–4%
Applicability to intended role (Relevance to COU)	Min 3; Max 5; Mean 4.2	Direct mapping for bare-st crush; strong mapping coupled bending